Credit Portfolio Governance Engine — LOD2 Report

1. Introduction & Opening Hook

In a global banking landscape shaped by macroeconomic volatility and the continuous tightening of Basel III and IV regulatory standards, credit risk management extends far beyond the ex-post accounting of losses. *"To govern is to foresee"*: this maxim resonates profoundly within the Second Line of Defense (LOD2), where control independence and the quantitative rigor of models serve as the ultimate guardians of institutional solvency.

2. Problem Statement

How can we design and industrialize a wholesale portfolio governance and monitoring engine capable of unifying—within a single ecosystem—the dynamic aggregation of credit and counterparty exposures, automated Legal Lending Limit (LLL) compliance control, and forward-looking Early Warning System (EWS) backtesting?

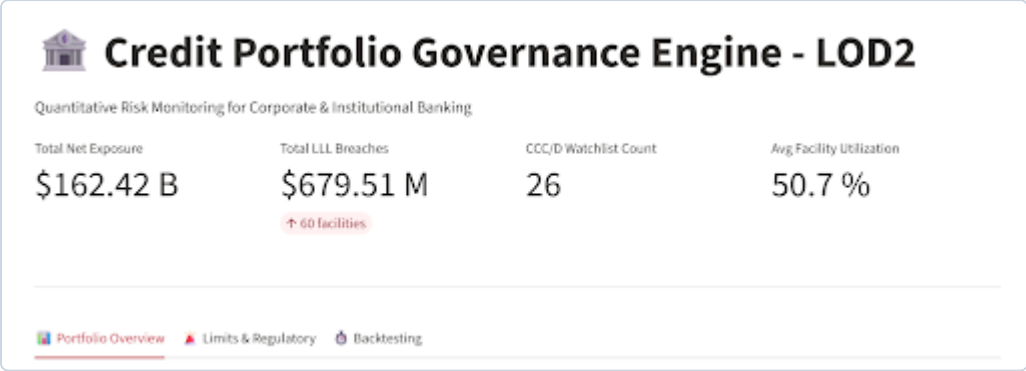


Figure 1: Synthetic Dashboard — Key Risk Indicators (KRIs) and Total Global Exposure

3. Analytical Development & Methodology

The project relies on a modular Python OOP architecture backed by a SQLite database. Consolidated net exposure integrates traditional credit risk and counterparty credit risk (CCR) via the formula: `Total_Exposure = Drawn_Amount + max(Trading_MtM, 0) + PFE`.

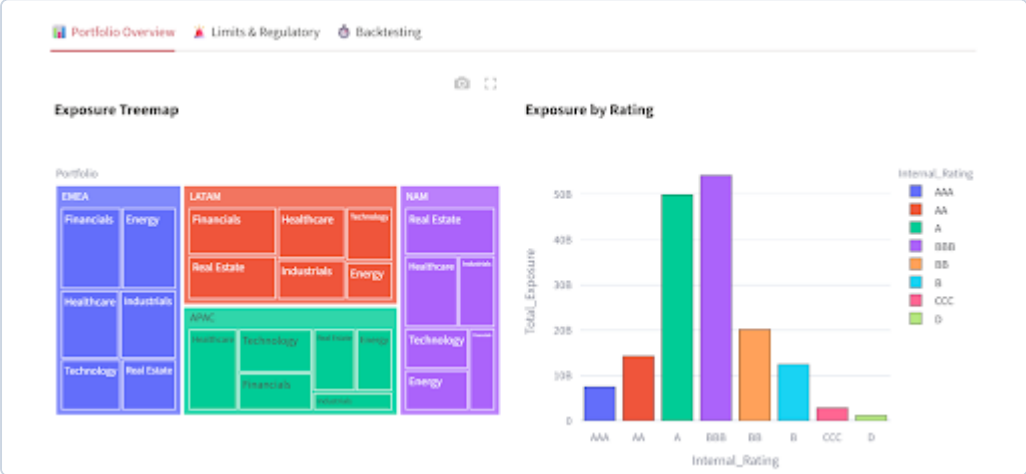


Figure 2: Sector and Geographic Portfolio Mapping (Treemap & Rating Distribution)

4. Results Interpretation

Analysis of the 500-counterparty portfolio reveals critical concentration pockets. As illustrated in *Figure 2*, geographic diversification (EMEA, NAM, APAC, LATAM) masks vulnerabilities within speculative rating tiers (BBB and below). The monitoring module (*Figure 3*) highlights Legal Lending Limit (LLL) breaches requiring immediate first-line corrective actions.

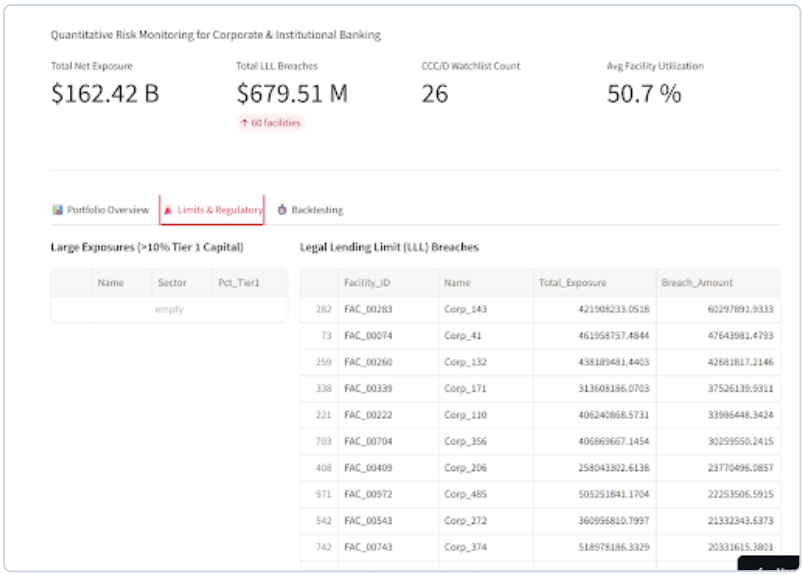


Figure 3: Legal Lending Limit (LLL) Breaches and Large Regulatory Exposures Ledger

5. Backtesting & Early Warning System (EWS) Validation

Time-series backtesting (*Figure 4*) validates the EWS model by intersecting credit rating downgrades to High Yield with facility utilization rates exceeding 85%, isolating systemic counterparties for Watchlist placement.

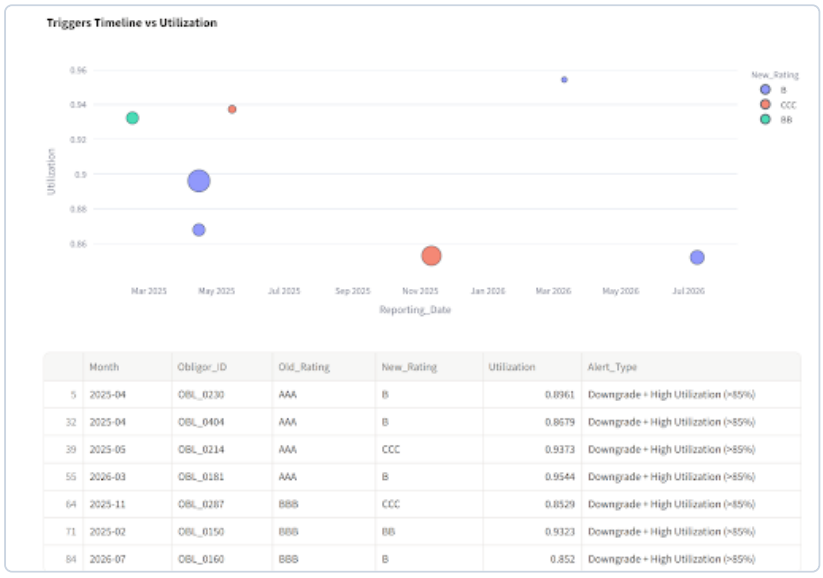


Figure 4: Early Warning Triggers Timeline Correlated with Potential Future Exposure (PFE)

6. Conclusion & Forward Outlook

In response to the initial problem statement, the **credit-portfolio-governance-engine** demonstrates that rigorous automation of credit risk monitoring significantly strengthens the second line of defense posture. By combining robust SQL foundations, object-oriented logic, and interactive Streamlit visualization, this project successfully addresses the complex equation of CIB regulatory compliance.

Forward Outlook: This engine serves as an evolutionary baseline. A natural extension involves integrating climate stress-testing scenarios (ESG factors) and deploying predictive Machine Learning models for advanced counterparty default anticipation.